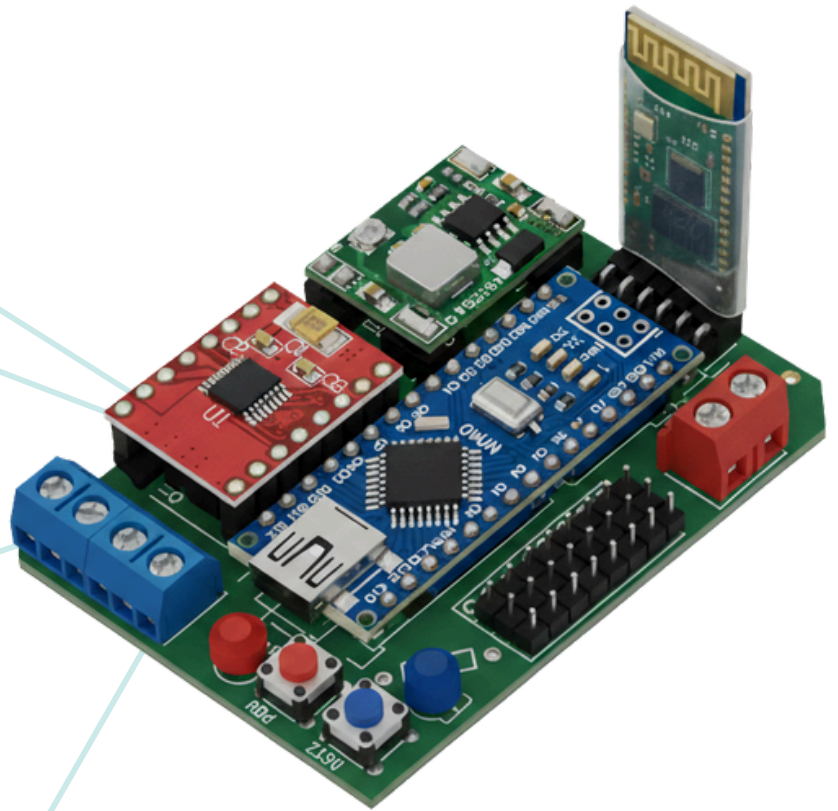


NOVA DYNAMIX

ONECHIP NANO SHIELD



USER MANUAL

Introduction

The ONECHIP Nano Shield is a compact, all-in-one Arduino Nano expansion board developed by Nova Dynamix.

It simplifies motor control, sensor interfacing, and wireless communication for robotics and embedded projects.

Whether you're a student, hobbyist, or educator, this shield provides a quick and reliable platform for experimenting with automation, IoT, and robotics. This manual guides you through features, connections, and usage with sample projects and workshops.



Technical Specifications

ONECHIP NANO SHIELD consists of:

- Microcontroller: Arduino Nano
- Motor Driver: TB6612FNG Dual DC Motor Driver
- Bluetooth Module: HC-05
- Buck Converter: MP1584
- Buttons: 2 Onboard Push Buttons
- LEDs: 2 Onboard LEDs
- Power Switch: 1 On/Off Toggle
- Available pins: 8 Analog + 2 Digital
- Input Voltage: 6–12V DC
- Output Voltage: 5V

Applications & Use Cases:

Robotics Competitions:

- Create competition-ready robots for line-following, obstacle avoidance, soccer, and sumo.

Innovation & Advanced Projects:

- Start with basic robots and develop advanced robotic systems using ONECHIP.

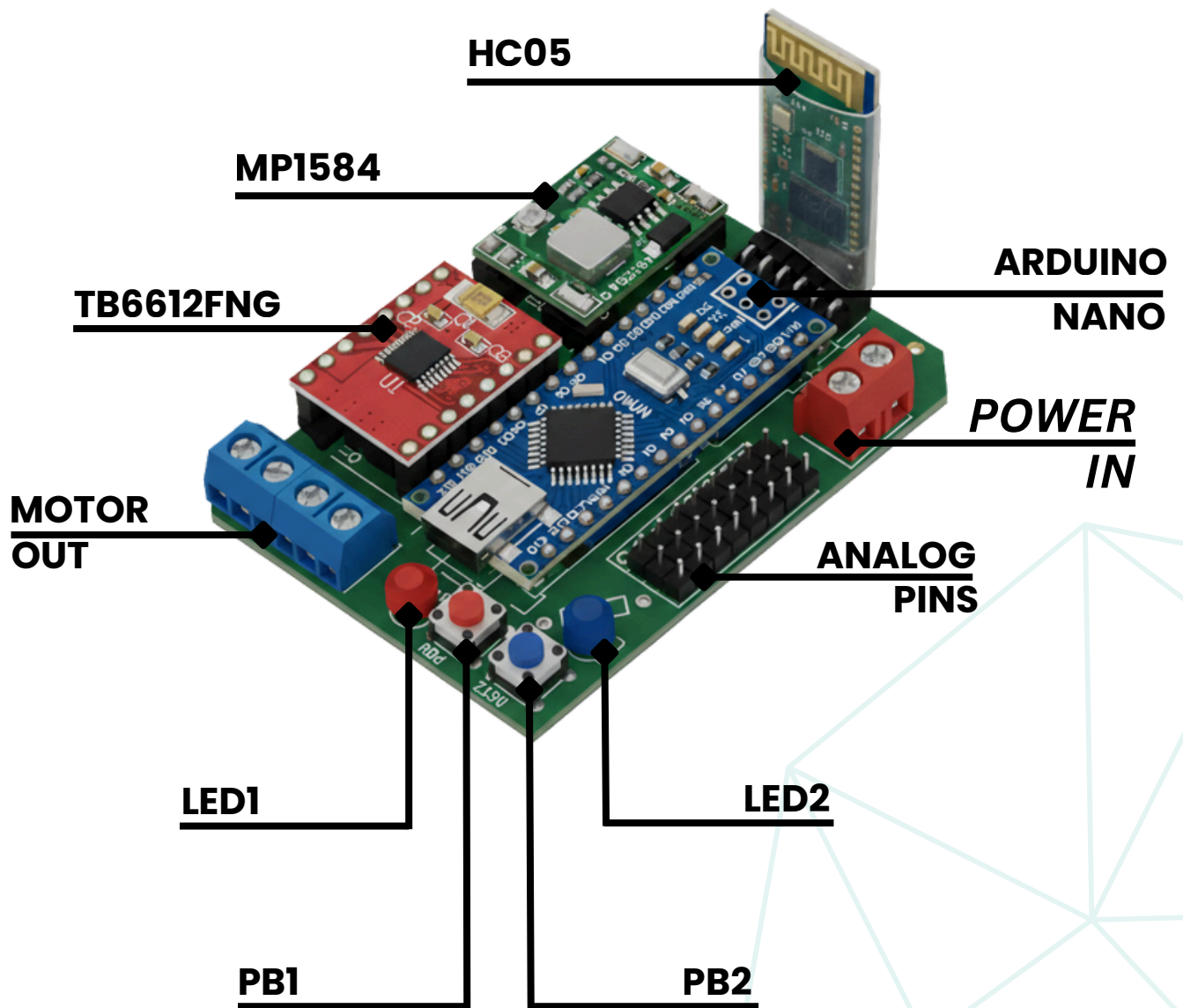
Academic & College Projects:

- Build prototypes and final-year projects while gaining hands-on experience in robotics and programming.

Advanced Robotics & IoT Learning:

- Explore autonomous navigation, sensor integration, wireless communication, and intelligent robotic systems.

Pinout and Layout Overview

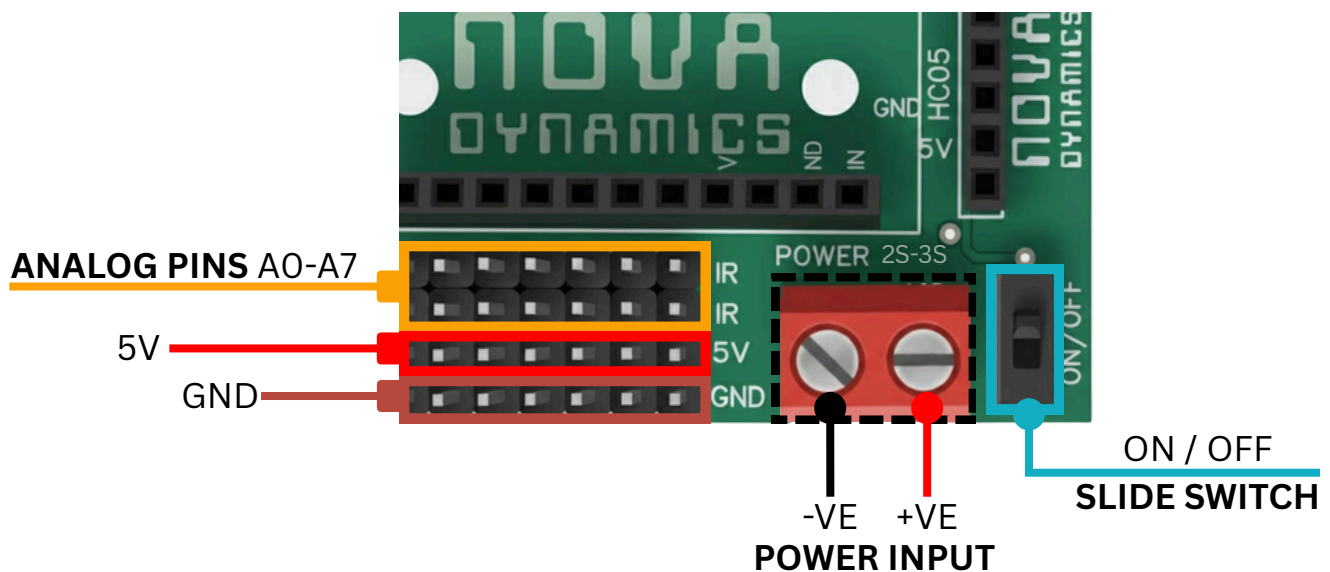


PINMAPPING

ARDUINO PIN	COMPONENT	STATUS
D0	HC05 Tx	Used
D1	HC05 Rx	Used
D2	Push Button 2	Used
D3	Push Button 1	Used
D4 & D5	—	Free
D6	TB6612FNG M2S	Used
D7	TB6612FNG M2A	Used
D8	TB6612FNG M2B	Used
D9	TB6612FNG M1A	Used
D10	TB6612FNG M1B	Used
D11	TB6612FNG M1S	Used
D12	LED 1	Used
D13	LED 2	Used
A0-A7	—	Free

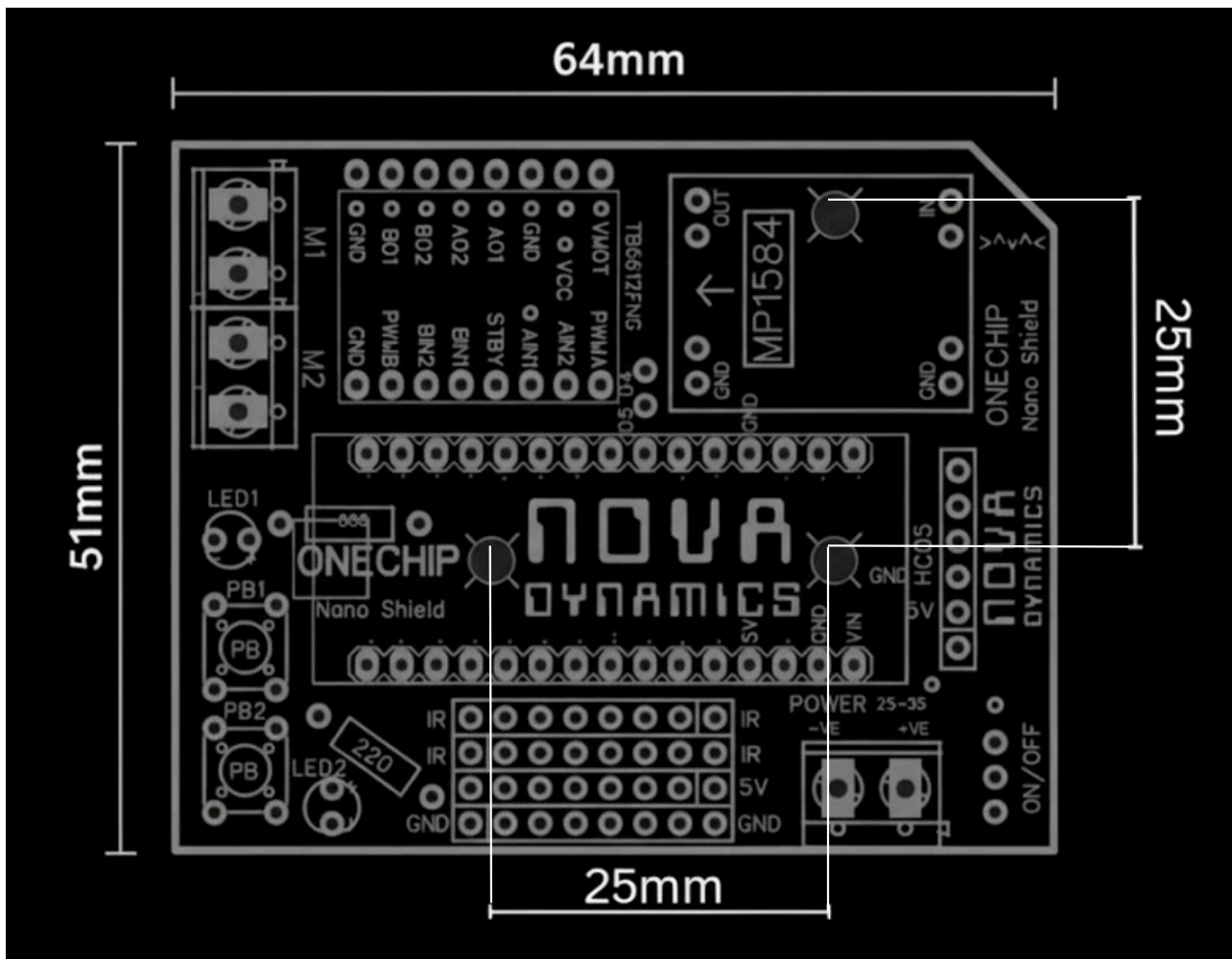
Power Connectivity

- Use a 6–12V DC input to power the shield.
- A 2S or 3S Li-ion battery pack is recommended for stable and efficient operation.
- The onboard MP1584 buck converter provides a regulated 5V supply to all components.
- Adjust the potentiometer on the buck converter to output exactly 5V before connecting sensors or modules.



- Power the motors only through the external 12V input – never from the USB connection.
- The USB port should be used only for programming or low-power testing.
- Check polarity carefully before connecting any external power source

PHYSICAL DIMENSIONS



- **Outer dimensions** are 64 mm x 51 mm
- **Mounting Holes** form an isosceles right-angled triangle with a side of 25mm

Component Description

MICROCONTROLLER

- The Arduino Nano is a compact microcontroller board based on the ATmega328P, featuring 14 digital I/O pins, 8 analog inputs, and 6 PWM outputs. It operates at 5V and provides a simple, reliable platform for controlling motors, sensors, communication modules, and other peripherals in compact robotics and embedded-system projects.

MOTOR DRIVER

- The TB6612FNG motor driver is a compact dual H-bridge driver that controls two DC motors independently with forward, reverse, and PWM-based speed control. It operates efficiently with minimal heat and provides up to 1.2A continuous current per channel, making it ideal for battery-powered robots and motor control projects.

BUCK CONVERTOR

- The MP1584 buck converter is a high-efficiency DC-DC step-down converter that reduces higher input voltages to a stable, adjustable output with up to 3A current capacity. Compact and reliable, it's commonly used to power 5V or 3.3V circuits in robotics and embedded systems.

ONECHIP LIBRARY

What is the OneChip Library?

The OneChip Library is a class-based Arduino library that simplifies control of the OneChip Nano Shield's components (motor driver, buttons, LEDs, ultrasonic sensor, IR receiver, PID controller) through a unified API.

Key features include:

- Independent control of two DC gear motors
- Easy management of onboard LEDs and buttons
- Ultrasonic distance measurement in one line
- Standard IR remote command reception
- Control of up to 6 servos from A0–A5 headers
- Built-in PID controller for various behaviors
- Wireless connection via HC-05 Bluetooth module

To install the library:

1. Download Onechip Nano Shield.zip from GitHub.
2. Open Arduino IDE.
3. Go to Sketch → Include Library → Add .ZIP Library...
4. Select the zip file
5. Restart the IDE. Example sketches will appear under File → Examples → Onechip Nano Shield.

TIP: Open *Documentation.ino* (under examples/Documentation/) any time you forget a function signature — it's the fastest built-in reference.

Troubleshooting & Safety

Troubleshooting Tips

No Power / Board Not Turning On :

- Check the power polarity and make sure the input voltage is between 6–12V DC.
- Verify the MP1584 output is set to 5V using a multimeter.

Motors Not Running :

- Confirm motor wires are connected properly to the MOTOR OUT terminals.
- Make sure the external power (12V) is connected; motors will not run on USB power.
- Check if PWM pins or control pins from the Arduino are correctly programmed.
- Inspect the TB6612FNG for overheating.

Bluetooth (HC-05) Not Pairing :

- Ensure the module's LED is blinking (indicating power).
- Check Bluetooth visibility on your device.
- Verify TX/RX pins are correctly connected (cross RX↔TX).

MP1584 Overheating or Output Unstable :

- Reduce input voltage if using near 12V with high current.
- Double-check the output voltage and load current.

LEDs Not Working :

- Check the Arduino code for correct digital pin mapping.

Safety Guidelines:

- Always set the buck converter output to 5V before powering any components.
- Never power the motors via USB – use the dedicated power input.
- Double-check polarity before connecting batteries or DC supplies.
- Do not exceed 12V input or 3A load current on the buck converter.
- Avoid short circuits by keeping metal tools or wires away from the board when powered.
- Use a common ground when connecting external modules or sensors.
- Disconnect power while uploading code or wiring new components.
- If any component gets unusually hot, immediately disconnect power and inspect connections.
- Store and operate the board in a dry, static-free environment.

Contact Us

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